

Class 5: Data Viz with ggplot2

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Background

There are many graphics systems in R for making plots and figures. These include so-called “*base R*” *graphics* like the `plot()` function and add on packages like **ggplot2**.

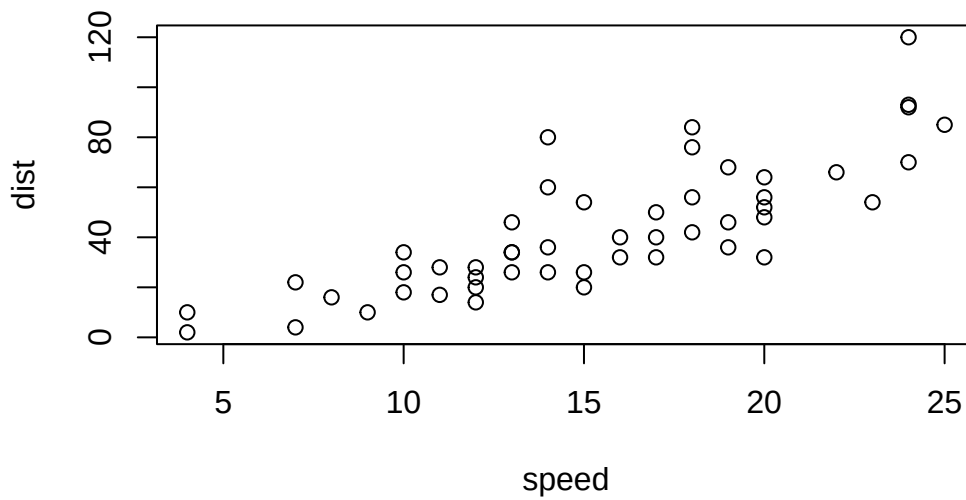
Let’s compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset:

```
head( cars )
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Before I can use `ggplot2` I need to install it on my computer. To do this we can use the function `install.packages("ggplot2")`

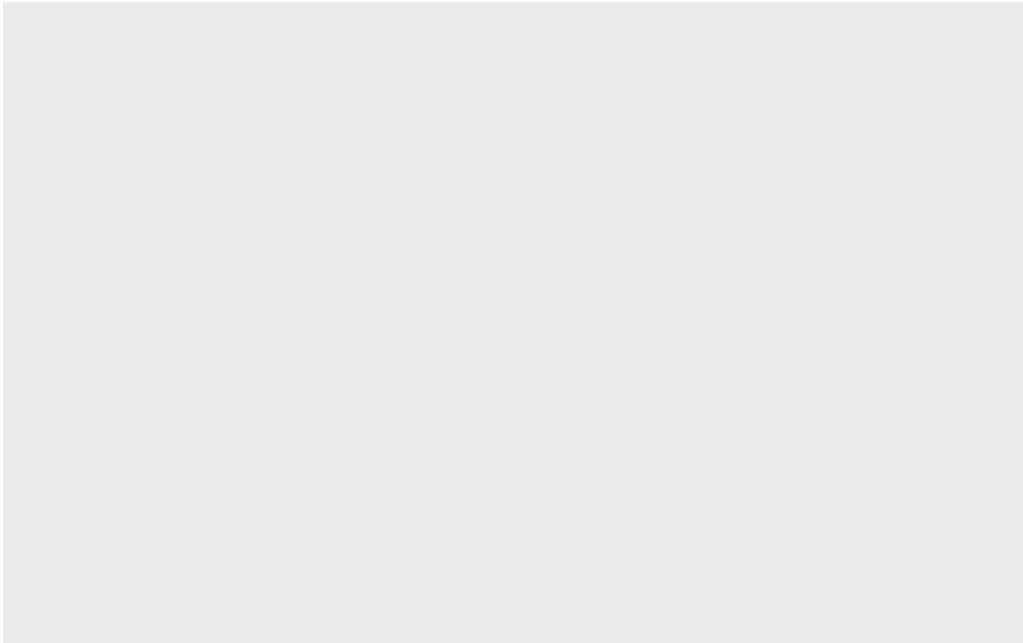
N.B. We never run `install.packages()` in our quarto doc (we run it once only in our console) as it would re-install the package every time we render our report.

Once installed we need to load up the package into our R brain:

```
library(ggplot2)
```

The main function in **ggplot2** package is called `ggplot()`

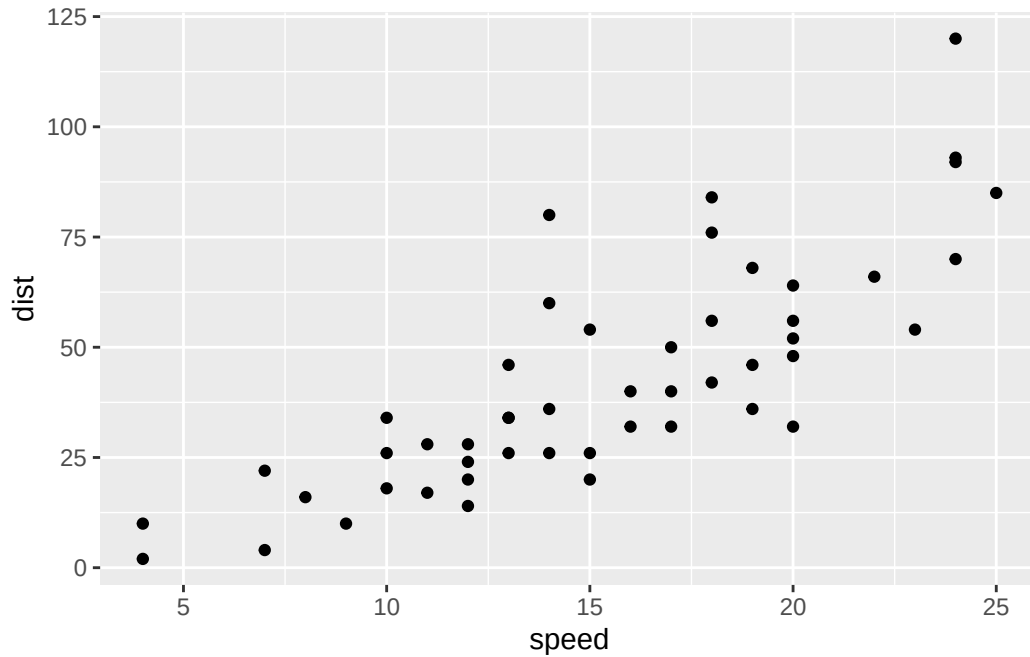
```
ggplot(cars)
```



Every ggplot has at least 3 layers:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot)
- the **geom** layer (how you want the plot drawn, e.g. points, lines, etc.)

```
ggplot(cars) +  
  aes(x=speed , y=dist) +  
  geom_point()
```

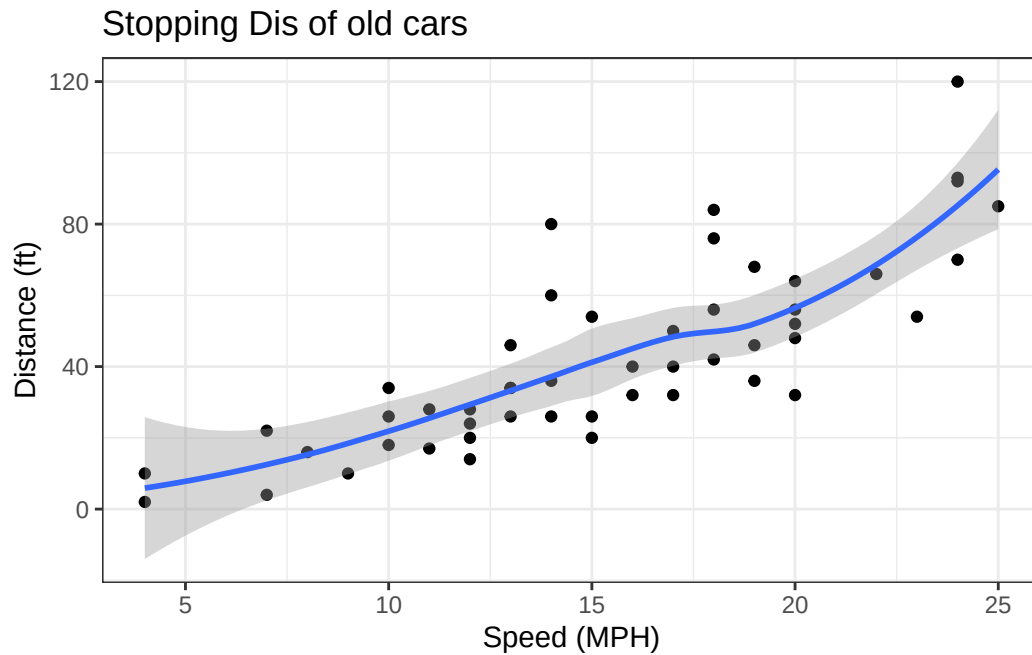


Add some custom features

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +  
  aes(x=speed , y=dist) +  
  geom_point() +  
  geom_smooth() +  
  theme_bw() +  
  labs(title="Stopping Dis of old cars",  
        x="Speed (MPH)",  
        y="Distance (ft)")
```

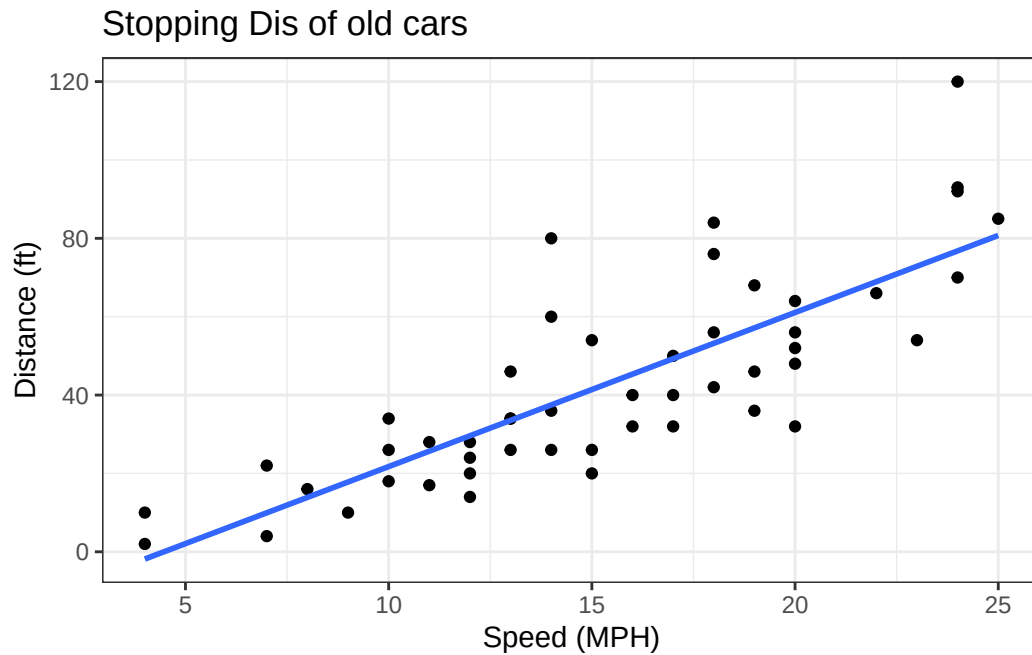
``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



Q. Can you make the `geom_smooth()` function produce a linear straight line fit to the data and turn off the “gray” error region?

```
ggplot(cars) +
  aes(x=speed , y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  theme_bw() +
  labs(title="Stopping Dis of old cars",
        x="Speed (MPH)",
        y="Distance (ft)")
```

`geom_smooth()` using formula = 'y ~ x'



Gene expression figure

Import the data to plot

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

Q. Use the `nrow()` function to find out how many genes are in this dataset. What is your answer?

```
nrow(genes)
```

```
[1] 5196
```

Q. Use the `colnames()` function and the `ncol()` function on the `genes` data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?

```
colnames(genes)
```

```
[1] "Gene"          "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

Q. How many upregulated genes are there?

```
sum(genes$State == "up")
```

```
[1] 127
```

A useful new function in this context is the `table()` function:

```
table(genes$State)
```

down	unchanging	up
72	4997	127

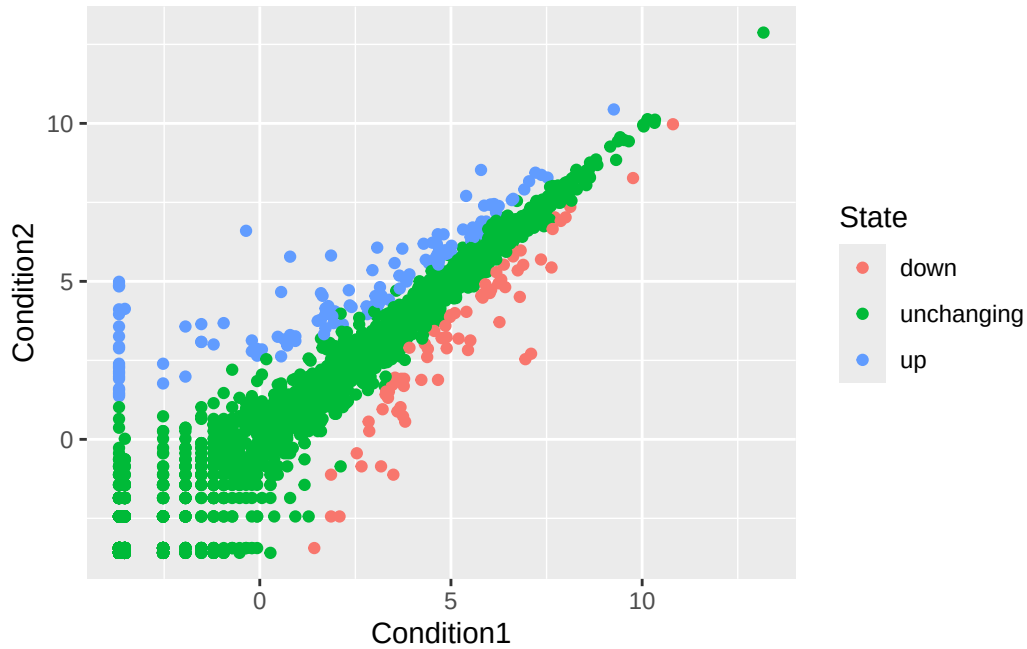
Q. Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

```
round(table(genes$State)/nrow(genes) * 100, 2)
```

down	unchanging	up
1.39	96.17	2.44

My first plot attempt

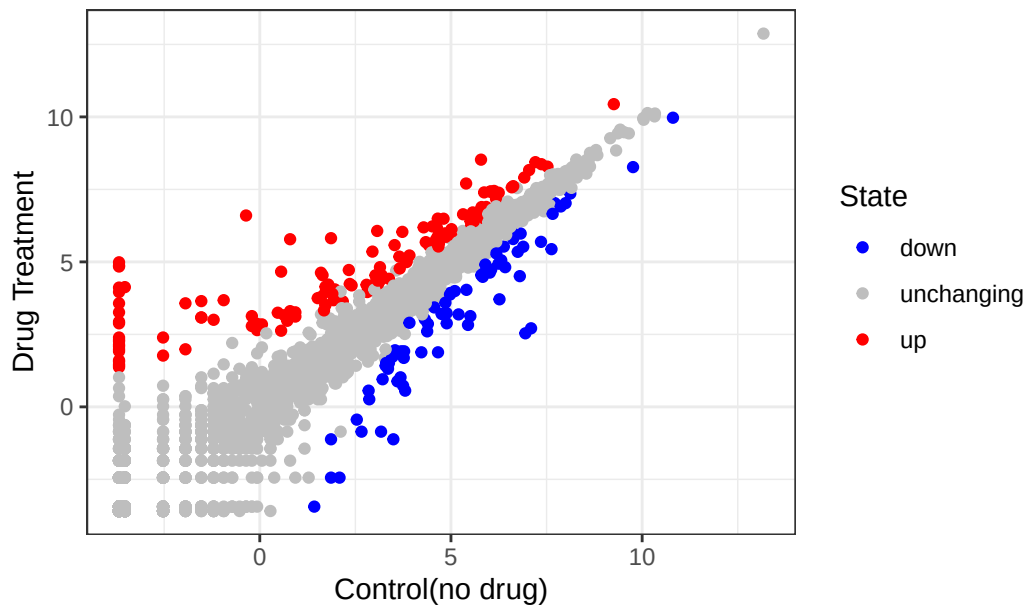
```
ggplot(genes) +
  aes(Condition1, Condition2, col=State) +
  geom_point()
```



Final differential expression analysis plot, with additional plot aesthetics:

```
ggplot(genes) +
  aes(Condition1, Condition2, col=State) +
  geom_point() + scale_colour_manual(values=c("blue","gray","red")) +
  theme_bw() +
  labs(title="Gene Expression Changes Upon Drug Treatment",
       x="Control(no drug)",
       y="Drug Treatment")
```


Gene Expression Changes Upon Drug Treatment



Going further

Here we read the famous gapminder dataset.

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"
gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. How many entries (i.e. rows) are in this dataset?

```
nrow(gapminder)
```

[1] 1704

Q. How many “country” entries are in this dataset?

```
length(table(gapminder$country))
```

[1] 142

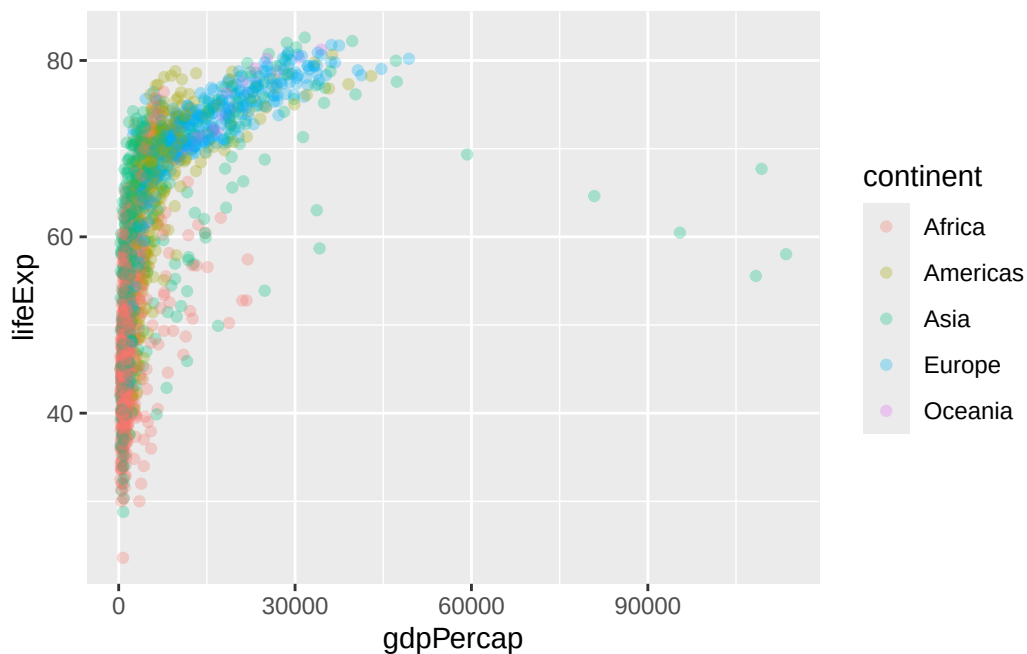
```
length(unique(gapminder$country))
```

[1] 142

Let’s make our first plot of the entire dataset:

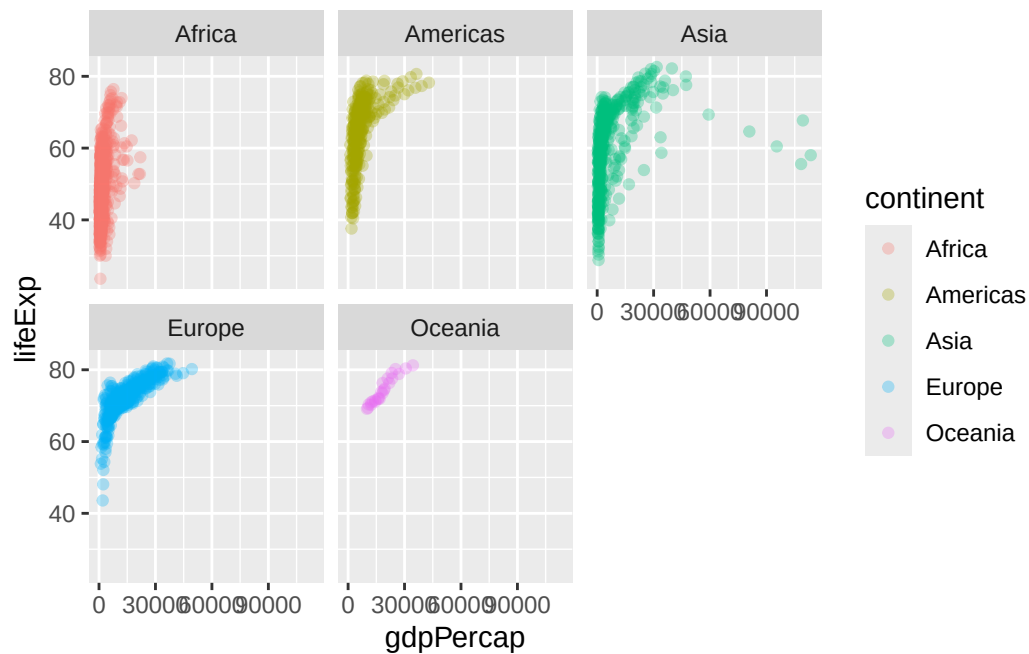
Plot of “gdpPercap” vs “lifeExp” colored by “continent”

```
p <- ggplot(gapminder) +  
  aes(gdpPercap, lifeExp, color=continent) +  
  geom_point(alpha=0.3)  
p
```



I can add more layers to p

```
p +  
  facet_wrap(~continent)
```



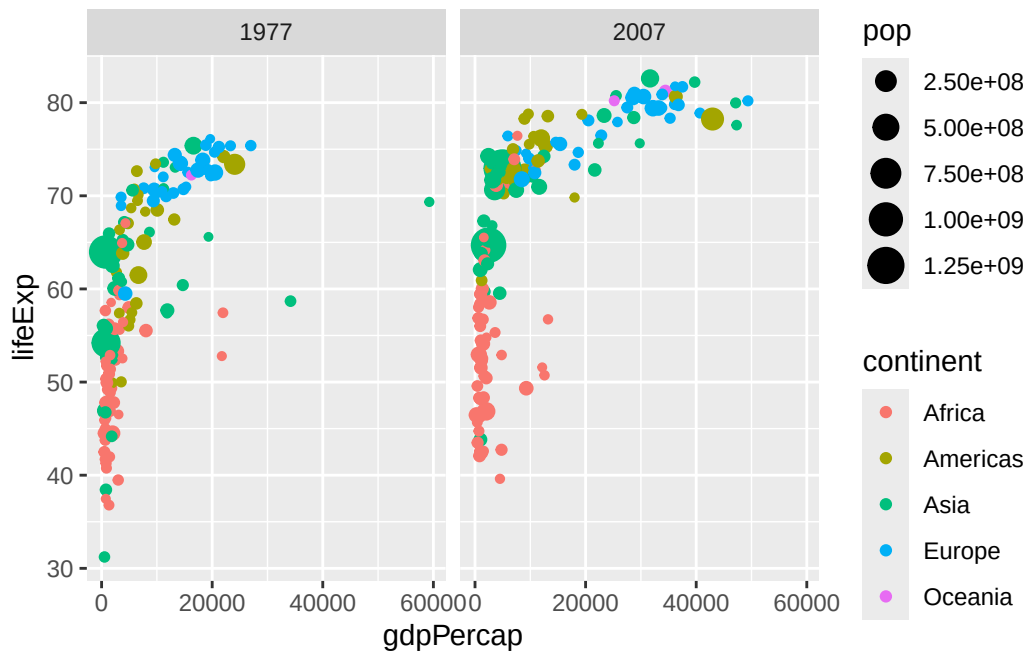
Make a plot for 1977 and 2007 only (not all the years in the dataset).

Q. First use the **dplyr** package and the **filter()** function from that package to extract the year 2007.

```
library(dplyr)
```

```
g07 <- filter(gapminder, year == 2007)  
g77 <- filter(gapminder, year == 1977)  
g <- filter(gapminder, year == 2007 | year == 1977)
```

```
ggplot(g) +  
  aes(gdpPercap, lifeExp, col=continent, size=pop) +  
  geom_point() +  
  facet_wrap(~year)
```

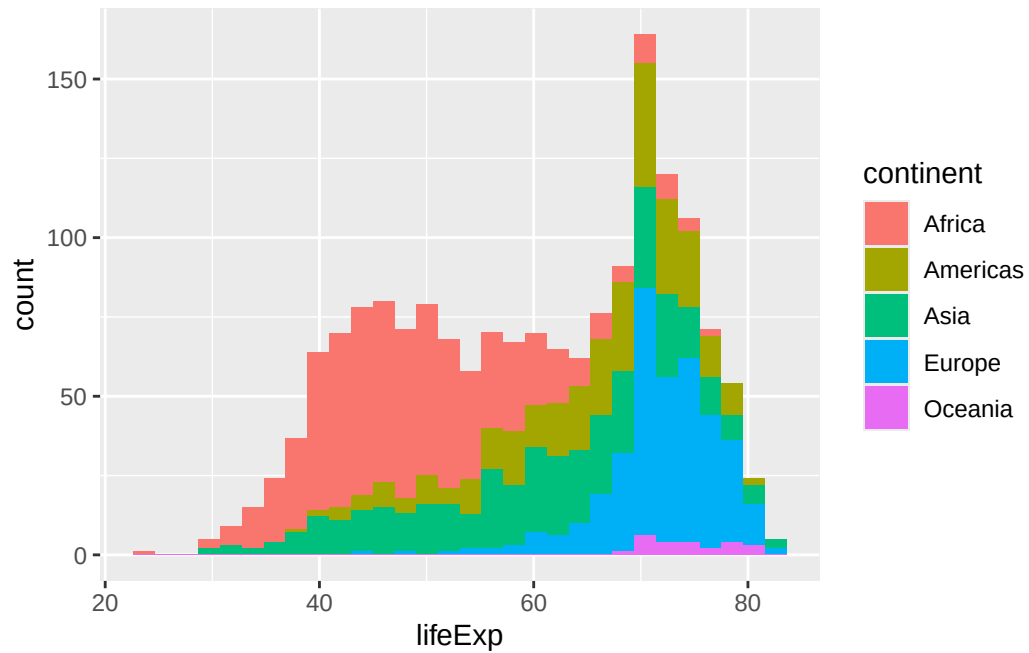


Q. Make a histogram of lifeExp colored by continent (use `fill=continent` or `col=continent`).

Q. Make a histogram of lifeExp faceted by continent.

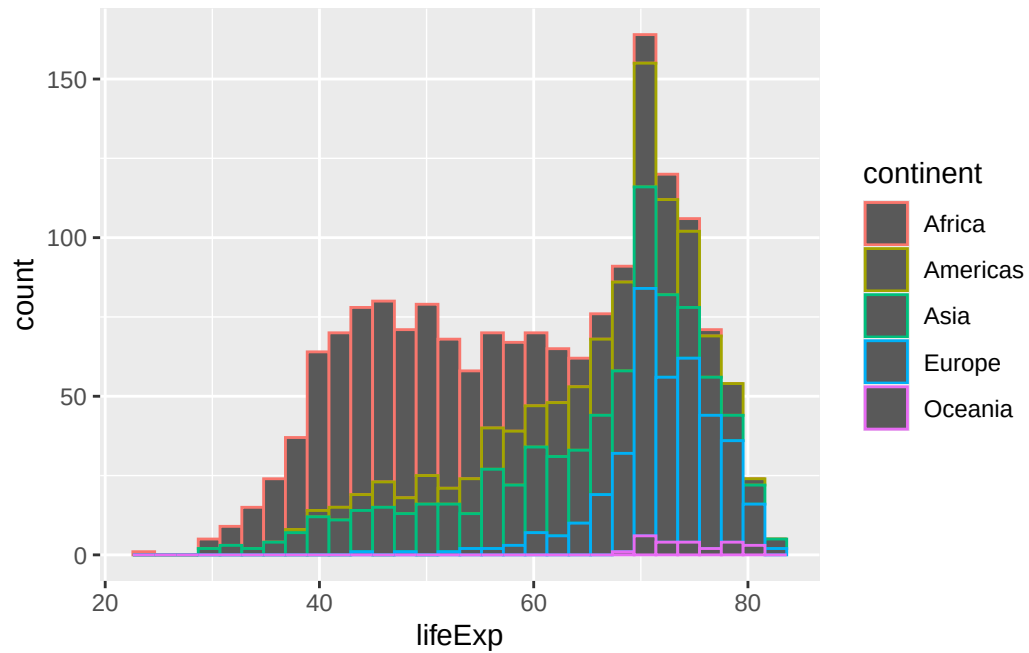
```
ggplot(gapminder) +
  aes(lifeExp, fill=continent) +
  geom_histogram()
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



```
ggplot(gapminder) +  
  aes(lifeExp, col=continent) +  
  geom_histogram()
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



```
ggplot(gapminder) +  
  aes(lifeExp) +  
  geom_histogram() +  
  facet_wrap(~continent)
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

